VICTORIA MARINE, BC, Canada

WMO: **712023**

Lat: **48.370N** Lon: **123.750W** Elev: **32** StdP: **100.94** Time Zone: **-8.00 (NAP)** Period: **82-92** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	12	-5.6	-2.6	-11.4	1.4	-3.5	-7.8	2.0	-0.5	16.3	3.0	14.2	3.9	6.7	90	0.632	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	8	6.8	21.1	15.1	19.3	14.4	17.8	13.7	15.8	20.0	14.9	18.2	14.2	17.0	2.9	270	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	14.2	10.2	16.6	13.3	9.5	15.9	12.6	9.1	15.3	44.4	20.4	41.8	18.3	39.8	17.3	24.1	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	13.5	11.7	9.8	DB	-7.2	26.1	3.6	0.9	-9.8	26.7	-11.9	27.3	-14.0	27.8	-16.6	28.4	(4)
(5)				WB	-8.3	19.1	3.3	2.9	-10.7	21.1	-12.6	22.9	-14.5	24.5	-16.9	26.6	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg		9.1	5.0	5.4	6.7	8.1	10.3	12.3	13.8	14.1	12.7	9.4	6.4	4.2	(6)
(7)		DBStd		4.13	2.62	3.13	1.89	1.92	1.92	1.65	1.57	1.54	1.99	2.27	3.23	3.17	(7)
(8)		HDD10.0		797	157	129	104	62	17	1	0	0	2	37	109	179	(8)
(9)		HDD18.3		3384	414	362	361	306	248	182	142	131	168	276	357	438	(9)
(10)		CDD10.0		455	1	1	1	6	27	69	117	127	84	19	3	0	(10)
(11)		CDD18.3		1	0	0	0	0	0	0	1	0	0	0	0	0	(11)
(12)		CDH23.3		11	0	0	0	0	1	2	5	1	1	0	0	0	(12)
(13)		CDH26.7		0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg		3.4	3.6	3.3	2.9	3.4	4.0	4.0	4.3	3.1	2.3	2.2	3.5	3.5	(14)
(15)	Precipitation	PrecAvg		1265	200	147	109	78	46	30	22	27	56	125	212	203	(15)
(16)		PrecMax		1633	344	269	219	174	105	94	89	117	122	351	443	356	(16)
(17)		PrecMin		832	24	56	61	27	6	6	1	0	2	25	56	12	(17)
(18)		PrecStd		228	88	65	45	38	25	20	20	27	38	85	101	84	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		12.5	12.8	13.5	17.3	19.9	22.4	23.8	22.3	22.1	17.2	13.4	11.1	(19)
(20)			MCWB		9.8	9.6	9.1	10.5	14.2	16.0	16.5	17.2	14.8	12.3	11.7	10.2	(20)
(21)		2%	DB		10.1	11.1	11.8	14.5	16.8	18.8	20.2	20.6	19.4	15.1	12.0	9.6	(21)
(22)			MCWB		8.5	8.9	8.6	10.0	12.4	14.2	14.8	15.6	14.2	12.7	10.6	8.2	(22)
(23)		5%	DB		9.3	10.1	10.7	12.9	14.8	16.4	18.1	19.0	17.8	13.8	11.1	8.9	(23)
(24)			MCWB		8.0	8.2	8.1	9.6	11.3	12.7	13.7	14.7	13.8	11.6	9.9	7.7	(24)
(25)		10%	DB		8.4	9.2	10.0	11.7	13.4	15.1	16.7	17.5	16.4	12.8	10.2	8.2	(25)
(26)			MCWB		7.2	7.7	7.7	8.9	10.6	11.9	13.1	13.9	13.2	10.9	9.0	7.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		10.6	10.5	10.6	11.5	14.5	15.8	16.5	18.6	16.6	14.3	12.2	10.4	(27)
(28)			MCDB		12.0	11.9	12.1	15.6	18.4	21.2	22.7	20.3	19.2	15.7	13.0	11.0	(28)
(29)		2%	WB		9.0	9.5	9.3	10.6	12.9	14.3	15.1	16.1	15.1	12.8	11.0	8.7	(29)
(30)			MCDB		9.6	10.5	10.8	13.6	16.4	18.1	19.5	19.8	18.0	14.5	11.5	9.3	(30)
(31)		5%	WB		8.2	8.7	8.7	9.9	11.7	13.2	14.2	15.0	14.2	12.0	10.1	8.0	(31)
(32)			MCDB		9.0	9.7	10.1	12.3	14.1	15.8	17.3	17.9	16.9	13.4	10.9	8.7	(32)
(33)		10%	WB		7.3	7.9	8.1	9.2	10.8	12.3	13.5	14.2	13.4	11.2	9.2	7.3	(33)
(34)			MCDB		8.2	9.0	9.5	11.2	12.9	14.6	16.1	16.8	16.0	12.5	10.0	8.1	(34)
(35)	Mean Daily Temperature Range		MDBR		4.1	4.9	5.9	6.0	5.5	5.8	6.3	6.8	7.0	5.6	4.2	4.1	(35)
(36)		5% DB	MCDBR		4.5	5.6	7.0	8.6	8.1	8.5	9.1	9.6	9.0	6.7	4.2	3.5	(36)
(37)			MCWBR		3.4	4.0	4.9	5.5	4.8	5.0	4.6	5.5	5.2	4.5	3.4	2.9	(37)
(38)		5% WB	MCDBR		3.9	4.8	5.9	7.5	7.3	7.7	8.1	8.6	8.0	5.7	3.7	3.2	(38)
(39)			MCWBR		3.4	3.8	4.5	5.1	4.7	5.0	4.7	5.4	5.2	4.1	3.3	2.9	(39)
(40)	Clear-Sky Solar Irradiance	taub			0.319	0.317	0.334	0.364	0.383	0.362	0.352	0.355	0.343	0.334	0.328	0.314	(40)
(41)		taud			2.509	2.542	2.504	2.409	2.376	2.451	2.522	2.511	2.544	2.554	2.487	2.485	(41)
(42)		Ebn at Noon			757	851	887	887	879	899	904	887	867	817	733	711	(42)
(43)		Edh at Noon			62	74	89	108	116	108	99	96	85	72	63	56	(43)
(44)	All-Sky Solar Radiation	RadAvg			0.93	1.82	2.81	4.26	5.41	5.91	6.37	5.37	3.90	2.17	1.12	0.76	(44)
(45)		RadStd			0.13	0.25	0.33	0.39	0.42	0.55	0.44	0.37	0.32	0.25	0.17	0.10	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (73 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page